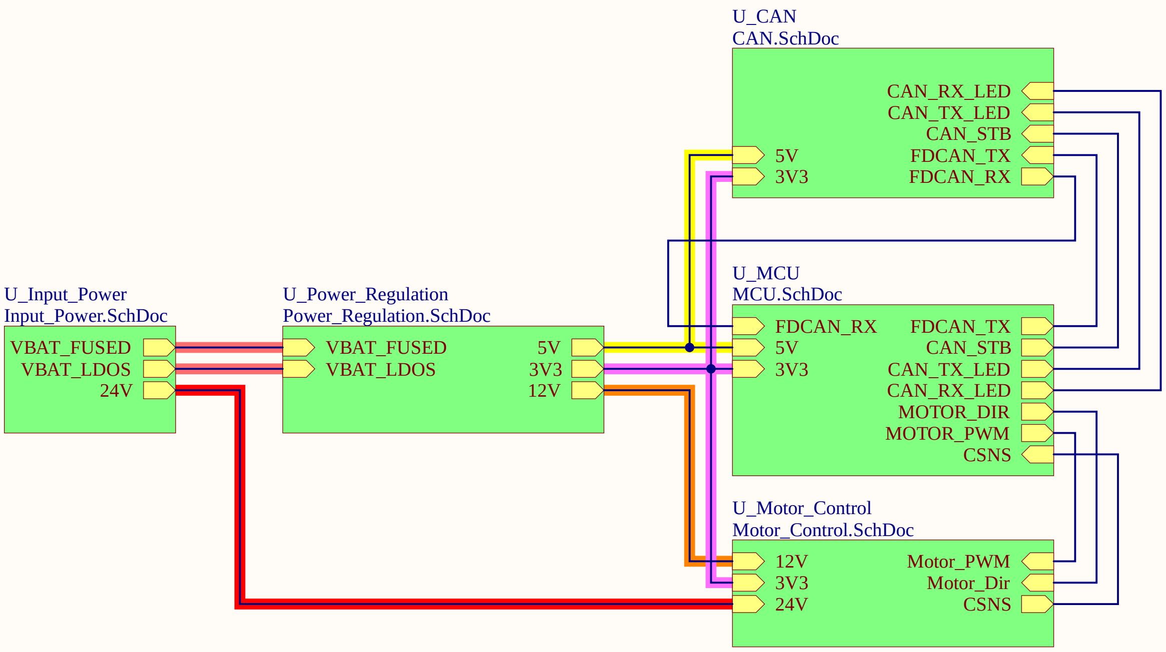
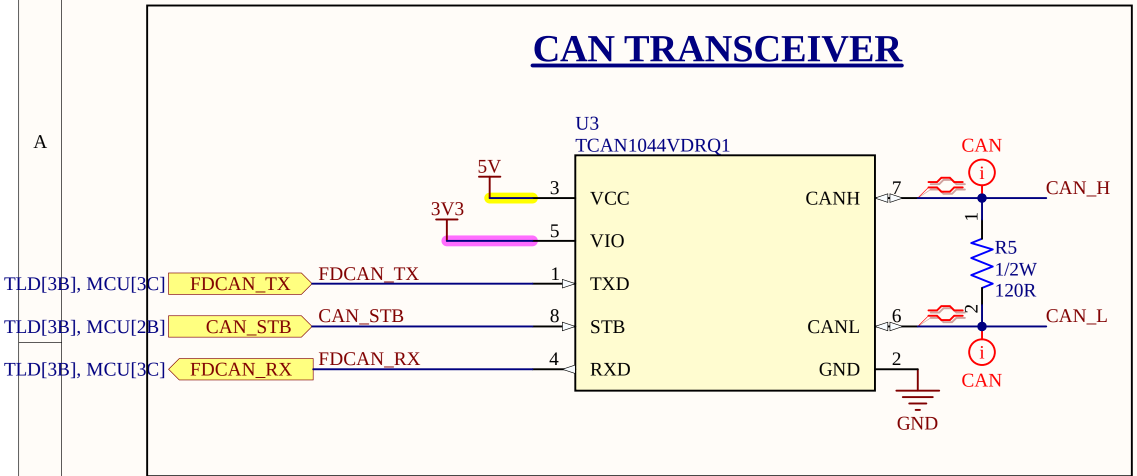
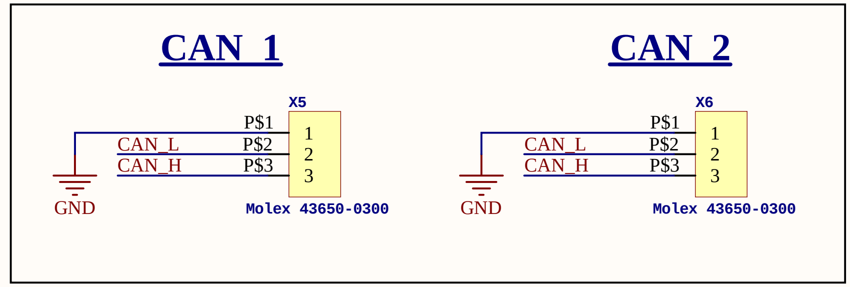
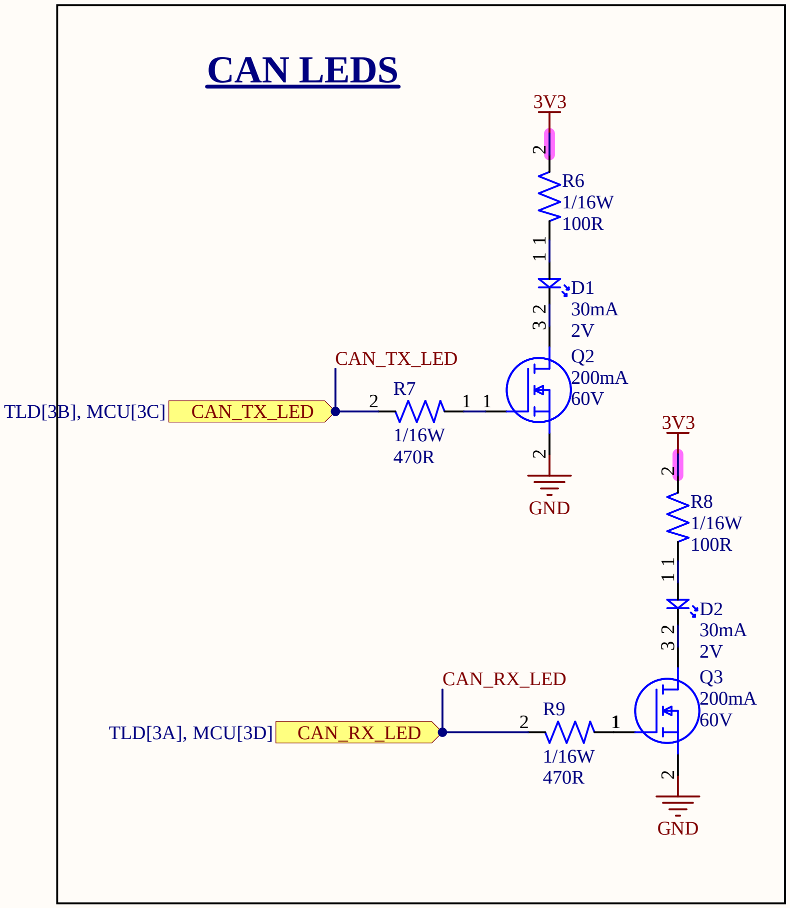
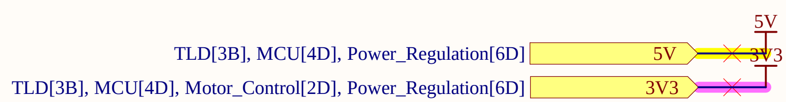

U_CAN
CAN.SchDoc
CAN_RX_LED
CAN_TX_LED
CAN_STB
5V
FDCAN_TX
3V3
FDCAN_RX
U_Input_Power
Input_Power.SchDoc
VBAT_FUSED
VBAT_LDOS
24V
U_Power_Regulation
Power_Regulation.SchDoc
VBAT_FUSED
VBAT_LDOS
5V
3V3
12V
U_MCU
MCU.SchDoc
FDCAN_RX
5V
3V3
FDCAN_TX
CAN_STB
CAN_TX_LED
CAN_RX_LED
MOTOR_DIR
MOTOR_PWM
CSNS
U_Motor_Control
Motor_Control.SchDoc
12V
3V3
24V
Motor_PWM
Motor_Dir
CSNS

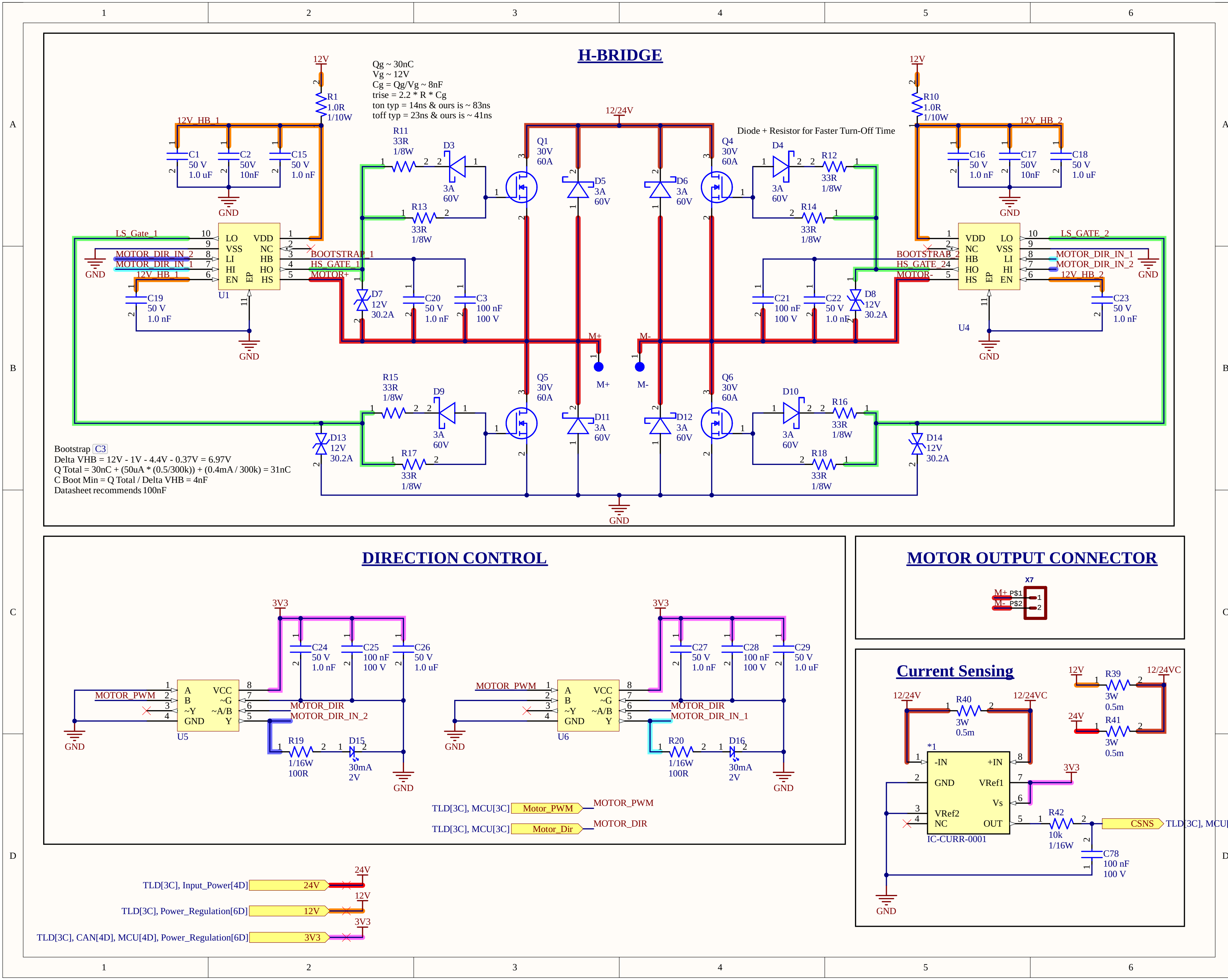




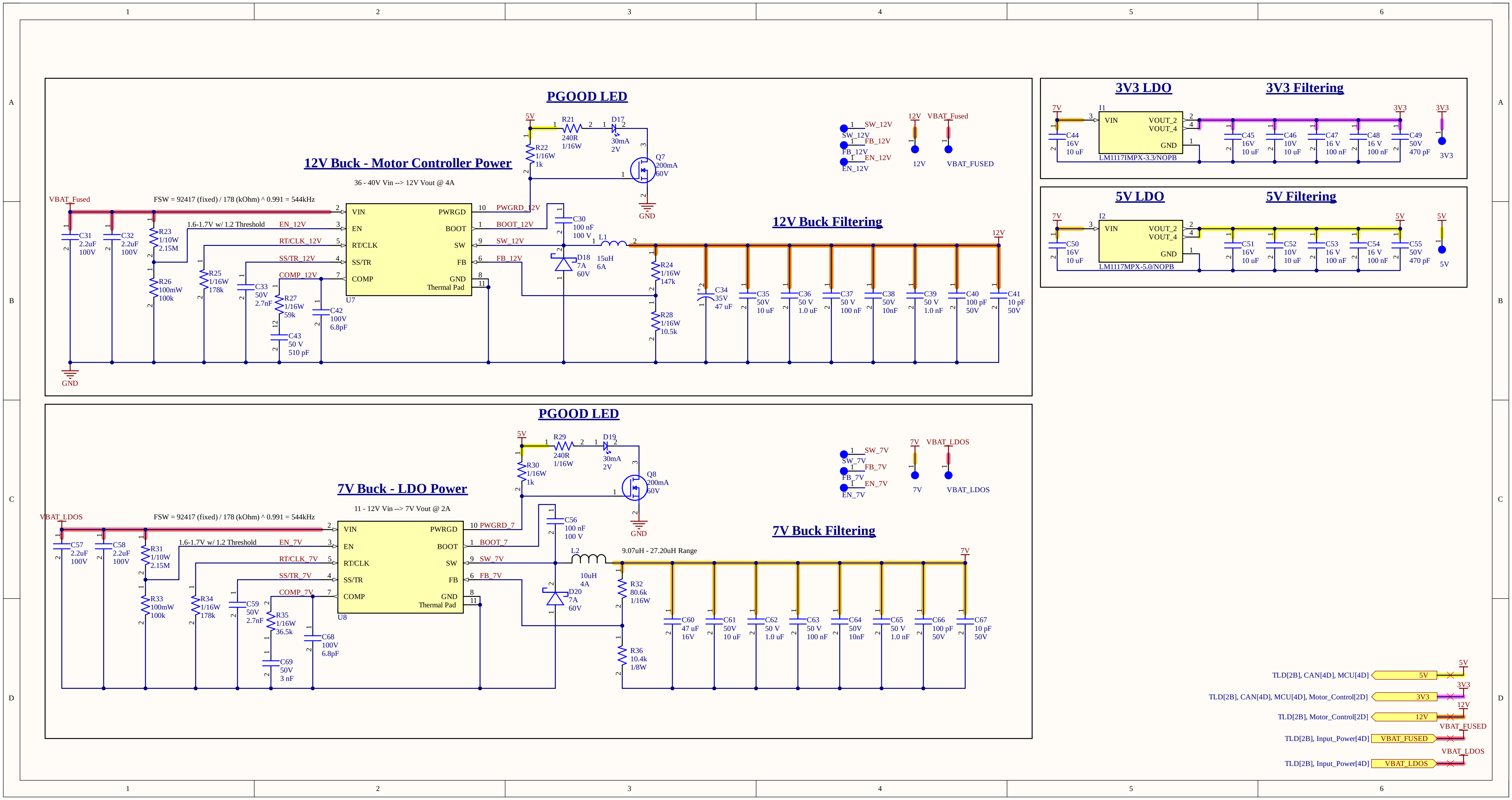


Pins		Type	Description
Name	No.		
TXD	1	Digital Input	CAN transmit data input
GND	2	GND	Ground connection
V_{CC}	3	Supply	5V supply voltage
RXD	4	Digital Output	CAN receive data output, tri-state when powered off
NC	5	—	No Connect (not internally connected); Devices without V_{IO}
V_{IO}		Supply	I/O supply voltage
CANL	6	Bus IO	Low-level CAN bus input/output line
CANH	7	Bus IO	High-level CAN bus input/output line
STB	8	Digital Input	Standby input for mode control, integrated pull up

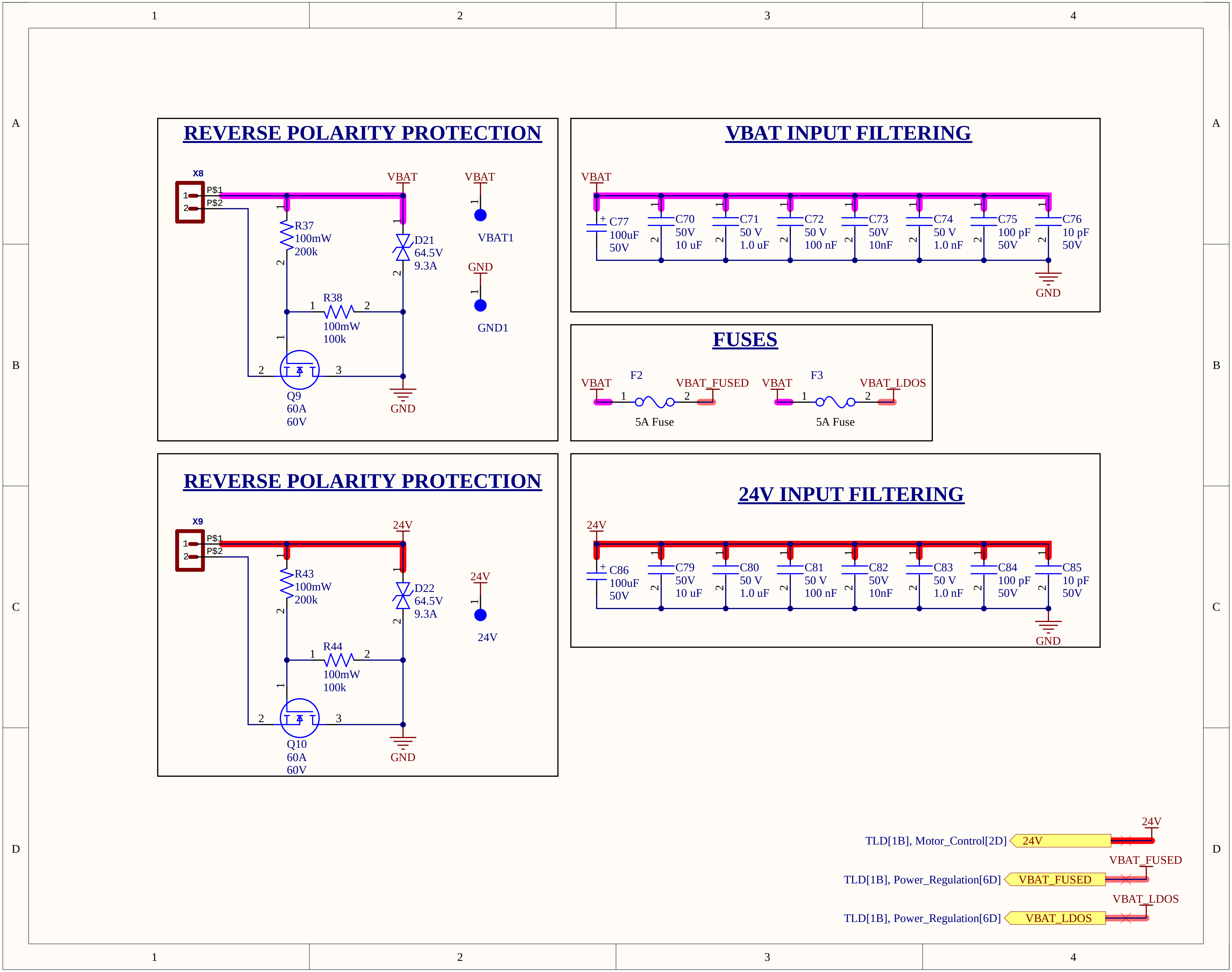


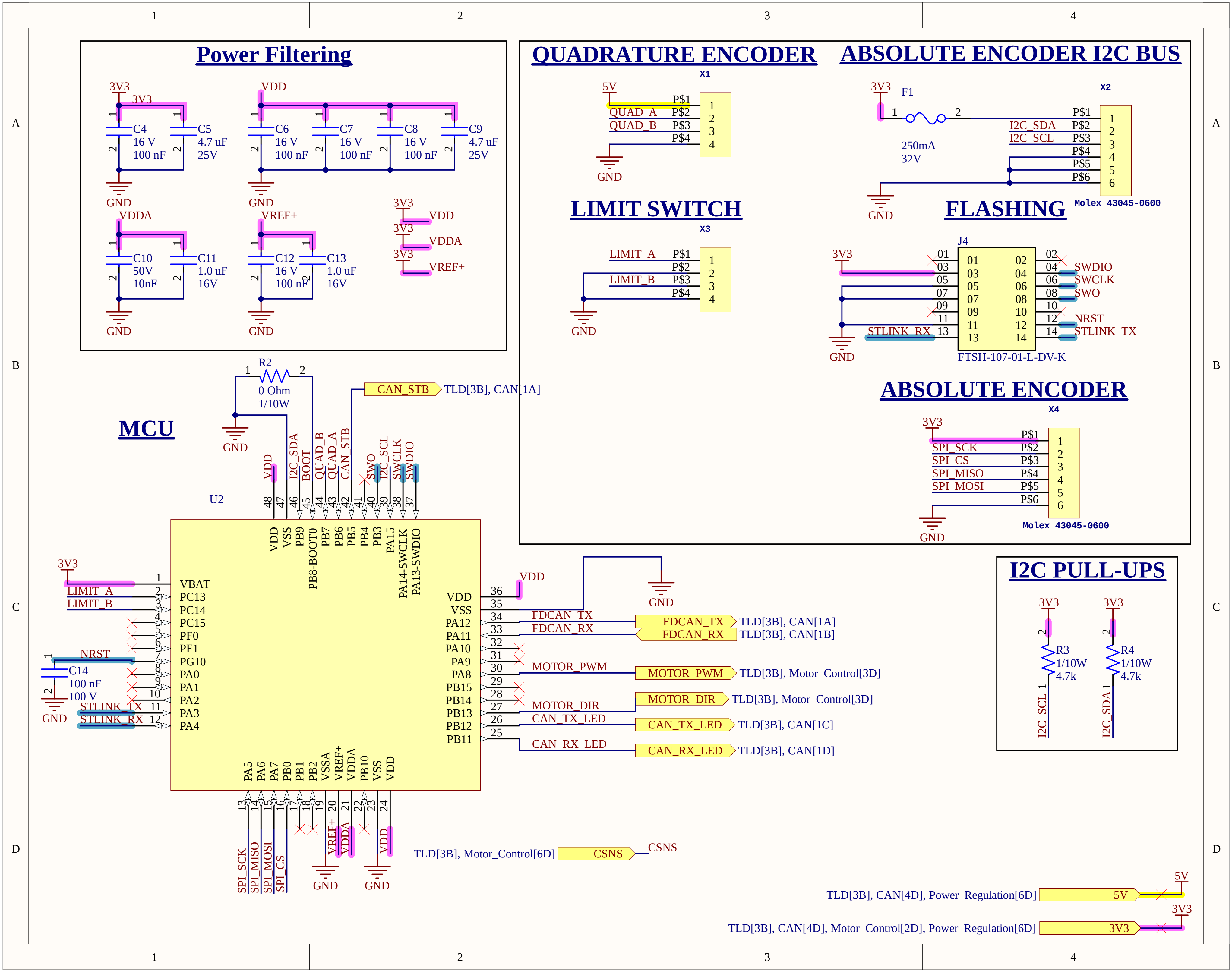


H-BRIDGE
Qg ~ 30nC
Vg ~ 12V
Cg = Qg/Vg ~ 8nF
trise = 2.2 * R * Cg
ton typ = 14ns & ours is ~ 83ns
toff typ = 23ns & ours is ~ 41ns
Diode + Resistor for Faster Turn-Off Time
12V
12/24V
12V_HB_1
12V_HB_2
R1 1.0R 1/10W
R10 1.0R 1/10W
C1 50 V 1.0 uF
C2 50V 10nF
C15 50 V 1.0 nF
C16 50 V 1.0 nF
C17 50V 10nF
C18 50 V 1.0 uF
R11 33R 1/8W
R12 33R 1/8W
R13 33R 1/8W
R14 33R 1/8W
D3 3A 60V
D4 3A 60V
Q1 30V 60A
Q4 30V 60A
D5 3A 60V
D6 3A 60V
LS_Gate_1
LS_GATE_2
MOTOR_DIR_IN_2
MOTOR_DIR_IN_1
BOOTSTRAP_1
BOOTSTRAP_2
HS_GATE_1
HS_GATE_24
MOTOR+
MOTOR-
LO VDD VSS NC LI HB HI HO EN EP HS
U1
U4
GND
C19 50 V 1.0 nF
C23 50 V 1.0 nF
D7 12V 30.2A
D8 12V 30.2A
C20 50 V 1.0 nF
C3 100 nF 100 V
C21 100 nF 100 V
C22 50 V 1.0 nF
M+
M-
R15 33R 1/8W
R16 33R 1/8W
D9 3A 60V
D10 3A 60V
Q5 30V 60A
Q6 30V 60A
D11 3A 60V
D12 3A 60V
D13 12V 30.2A
D14 12V 30.2A
R17 33R 1/8W
R18 33R 1/8W
Bootstrap C3
Delta VHB = 12V - 1V - 4.4V - 0.37V = 6.97V
Q Total = 30nC + (50uA * (0.5/300k)) + (0.4mA / 300k) = 31nC
C Boot Min = Q Total / Delta VHB = 4nF
Datasheet recommends 100nF
DIRECTION CONTROL
3V3
C24 50 V 1.0 nF
C25 100 nF 100 V
C26 50 V 1.0 uF
C27 50 V 1.0 nF
C28 100 nF 100 V
C29 50 V 1.0 uF
MOTOR_PWM
A B ~Y GND VCC ~G ~A/B Y
U5
U6
MOTOR_DIR
MOTOR_DIR_IN_2
MOTOR_DIR_IN_1
R19 1/16W 100R
R20 1/16W 100R
D15 30mA 2V
D16 30mA 2V
TLD[3C], MCU[3C] Motor_PWM MOTOR_PWM
TLD[3C], MCU[3C] Motor_Dir MOTOR_DIR
MOTOR OUTPUT CONNECTOR
X7
M+ P$1 1
M- P$2 2
Current Sensing
12V
24V
12/24VC
R39 3W 0.5m
R41 3W 0.5m
12/24V
R40 3W 0.5m
*1
-IN +IN GND VRef1 Vs VRef2 NC OUT
IC-CURR-0001
R42 10k 1/16W
CSNS TLD[3C], MCU[2D]
C78 100 nF 100 V
TLD[3C], Input_Power[4D] 24V
TLD[3C], Power_Regulation[6D] 12V
TLD[3C], CAN[4D], MCU[4D], Power_Regulation[6D] 3V3



PGOOD LED
12V Buck - Motor Controller Power
36 - 40V Vin --> 12V Vout @ 4A
FSW = 92417 (fixed) / 178 (kOhm) ^ 0.991 = 544kHz
1.6-1.7V w/ 1.2 Threshold
VBAT_Fused
GND
5V
R21 240R 1/16W
D17 30mA 2V
R22 1/16W 1k
Q7 200mA 60V
U7
VIN
EN
RT/CLK
SS/TR
COMP
PWRGD
BOOT
SW
FB
GND
Thermal Pad
PWRGD_12V
BOOT_12V
SW_12V
FB_12V
EN_12V
RT/CLK_12V
SS/TR_12V
COMP_12V
C30 100 nF 100 V
L1 15uH 6A
D18 7A 60V
R24 1/16W 147k
R28 1/16W 10.5k
C31 2.2uF 100V
C32 2.2uF 100V
R23 1/10W 2.15M
R26 100mW 100k
R25 1/16W 178k
C33 50V 2.7nF
R27 1/16W 59k
C43 50 V 510 pF
C42 100V 6.8pF
12V Buck Filtering
12V
C34 35V 47 uF
C35 50V 10 uF
C36 50 V 1.0 uF
C37 50 V 100 nF
C38 50V 10nF
C39 50 V 1.0 nF
C40 100 pF 50V
C41 10 pF 50V
VBAT_FUSED
3V3 LDO
3V3 Filtering
7V
I1
LM1117IMPX-3.3/NOPB
VOUT_2
VOUT_4
C44 16V 10 uF
C45 16V 10 uF
C46 10V 10 uF
C47 16 V 100 nF
C48 16 V 100 nF
C49 50V 470 pF
3V3
5V LDO
5V Filtering
I2
LM1117MPX-5.0/NOPB
C50 16V 10 uF
C51 16V 10 uF
C52 10V 10 uF
C53 16 V 100 nF
C54 16 V 100 nF
C55 50V 470 pF
5V
PGOOD LED
7V Buck - LDO Power
11 - 12V Vin --> 7V Vout @ 2A
FSW = 92417 (fixed) / 178 (kOhm) ^ 0.991 = 544kHz
1.6-1.7V w/ 1.2 Threshold
VBAT_LDOS
R29 240R 1/16W
D19 30mA 2V
R30 1/16W 1k
Q8 200mA 60V
U8
PWGRD_7
BOOT_7
SW_7V
FB_7V
EN_7V
RT/CLK_7V
SS/TR_7V
COMP_7V
C56 100 nF 100 V
L2 10uH 4A
9.07uH - 27.20uH Range
D20 7A 60V
R32 80.6k 1/16W
R36 10.4k 1/8W
C57 2.2uF 100V
C58 2.2uF 100V
R31 1/10W 2.15M
R33 100mW 100k
R34 1/16W 178k
C59 50V 2.7nF
R35 1/16W 36.5k
C69 50V 3 nF
C68 100V 6.8pF
7V Buck Filtering
C60 47 uF 16V
C61 50V 10 uF
C62 50 V 1.0 uF
C63 50 V 100 nF
C64 50V 10nF
C65 50 V 1.0 nF
C66 100 pF 50V
C67 10 pF 50V
TLD[2B], CAN[4D], MCU[4D] 5V
TLD[2B], CAN[4D], MCU[4D], Motor_Control[2D] 3V3
TLD[2B], Motor_Control[2D] 12V
TLD[2B], Input_Power[4D] VBAT_FUSED
TLD[2B], Input_Power[4D] VBAT_LDOS


REVERSE POLARITY PROTECTION
X8
P$1
P$2
VBAT
VBAT
VBAT1
R37
100mW
200k
D21
64.5V
9.3A
GND
R38
100mW
100k
GND1
Q9
60A
60V
GND
VBAT INPUT FILTERING
VBAT
C77
100uF
50V
C70
50V
10 uF
C71
50 V
1.0 uF
C72
50 V
100 nF
C73
50V
10nF
C74
50 V
1.0 nF
C75
100 pF
50V
C76
10 pF
50V
GND
FUSES
VBAT
F2
VBAT_FUSED
5A Fuse
VBAT
F3
VBAT_LDOS
5A Fuse
REVERSE POLARITY PROTECTION
X9
P$1
P$2
24V
R43
100mW
200k
D22
64.5V
9.3A
24V
24V
R44
100mW
100k
Q10
60A
60V
GND
24V INPUT FILTERING
24V
C86
100uF
50V
C79
50V
10 uF
C80
50 V
1.0 uF
C81
50 V
100 nF
C82
50V
10nF
C83
50 V
1.0 nF
C84
100 pF
50V
C85
10 pF
50V
GND
TLD[1B], Motor_Control[2D]
24V
24V
TLD[1B], Power_Regulation[6D]
VBAT_FUSED
VBAT_FUSED
TLD[1B], Power_Regulation[6D]
VBAT_LDOS
VBAT_LDOS
1
2
3
4
A
B
C
D


Power Filtering
QUADRATURE ENCODER
ABSOLUTE ENCODER I2C BUS
LIMIT SWITCH
FLASHING
ABSOLUTE ENCODER
MCU
I2C PULL-UPS
C4 16 V 100 nF
C5 4.7 uF 25V
C6 16 V 100 nF
C7 16 V 100 nF
C8 16 V 100 nF
C9 4.7 uF 25V
C10 50V 10nF
C11 1.0 uF 16V
C12 16 V 100 nF
C13 1.0 uF 16V
X1
X2
X3
X4
F1 250mA 32V
Molex 43045-0600
J4
FTSH-107-01-L-DV-K
R2 0 Ohm 1/10W
U2
C14 100 nF 100 V
R3 1/10W 4.7k
R4 1/10W 4.7k
CAN_STB TLD[3B], CAN[1A]
FDCAN_TX TLD[3B], CAN[1A]
FDCAN_RX TLD[3B], CAN[1B]
MOTOR_PWM TLD[3B], Motor_Control[3D]
MOTOR_DIR TLD[3B], Motor_Control[3D]
CAN_TX_LED TLD[3B], CAN[1C]
CAN_RX_LED TLD[3B], CAN[1D]
TLD[3B], Motor_Control[6D] CSNS
TLD[3B], CAN[4D], Power_Regulation[6D] 5V
TLD[3B], CAN[4D], Motor_Control[2D], Power_Regulation[6D] 3V3

Comment	Description	Designator	Footprint	LibRef	Quantity
IC-CURR-0001	Current Sense Amplifier 1 Circuit 8-MSOP	* 1	IC-CURR	IC-CURR-0001	1
5003	Oranange PC Test Point, Miniature Phoisphor Bronze Silver Plating Through Hole Mounting Type	3V3, 5V, 7V, 12V, 24V, EN_7V, EN_12V, FB_7V, FB_12V, GND1, M-, M+, SW_7V, SW_12V, VBAT1, VBAT_FUSED, VBAT_LDOS	TEST-POINT-0001	TEST-POINT-0001	17
CC0805KKX7R9BB105	CAP CER 1UF 50V X7R 0805	C1, C18, C26, C29, C36, C62, C71, C80	CAP-0805	CAP-1U-0001	8
GRM2195C1H103JA01D	CAP CER 10000PF 50V C0G/NP0 0805	C2, C10, C17, C38, C64, C73, C82	CAP-0805	CAP-10N-0001	7
CC0603KRX7R0BB104	CAP CER 0.1UF 100V X7R 0603	C3, C14, C21, C25, C28, C30, C56, C78	CAP-0603	CAP-100N-0001	8
CC0402KRX7R7BB104	CAP CER 0.1UF 16V X7R 0402	C4, C6, C7, C8, C12, C47, C48, C53, C54	CAP-0402	CAP-0.1U-0001	9
CC0603KRX5R8BB475	CAP CER 4.7UF 25V X5R 0603	C5, C9	CAP-0603	CAP-4.7U-0001	2
CC0603KRX7R7BB105	CAP CER 1UF 16V X7R 0603	C11, C13	CAP-0603	CAP-1U-0002	2
CC0402KRX7R9BB102	CAP CER 1000PF 50V X7R 0402	C15, C16, C19, C20, C22, C23, C24, C27, C39, C65, C74, C83	CAP-0402	CAP-1N-0001	12
CC1206KKX7R0BB225	CAP CER 2.2UF 100V X7R 1206	C31, C32, C57, C58	CAP-1206	CAP-2.2U-0001	4
GRM2165C1H272JA01D	CAP CER 2700PF 50V C0G/NP0 0805	C33, C59	CAP-0805	CAP-2.7N-0001	2
EEH-ZC1V470P	CAP ALUM HYBRID 47UF 20% 35V SMD	C34	CAP-Radial-SMD	CAP-47U-0002	1
CL32B106KBJNNNE	CAP CER 10UF 50V X7R 1210	C35, C61, C70, C79	CAP-1210	CAP-10U-0002	4
CC0805KRX7R9BB104	CAP CER 0.1UF 50V X7R 0805	C37, C63, C72, C81	CAP-0805	CAP-0.1U-0002	4
CC0402JRNPO9BN101	CAP CER 100PF 50V C0G/NPO 0402	C40, C66, C75, C84	CAP-0402	CAP-100P-0001	4
CC0402JRNPO9BN100	CAP CER 10PF 50V C0G/NPO 0402	C41, C67, C76, C85	CAP-0402	CAP-10P-0001	4
CC0603CRNPO0BN6R8	CAP CER 6.8PF 100V C0G/NPO 0603	C42, C68	CAP-0603	CAP-6.8P-0001	2
0402N511J500CT	CAP CER 510PF 50V C0G/NP0 0402	C43	CAP-0402	CAP-510P-0001	1
293D106X9016A2TE3	CAP TANT 10UF 10% 16V 1206	C44, C45, C50, C51	CAP-1206	CAP-10U-0004	4
CC0603KRX5R6BB106	CAP CER 10UF 10V X5R 0603	C46, C52	CAP-0603	CAP-10U-0001	2
CC0805JRNPO9BN471	CAP CER 470PF 50V C0G/NPO 0805	C49, C55	CAP-0805	CAP-470P-0001	2
C3225X7R1N106K250AC	CAP CER 47UF 16V X5R 1210	C60	CAP-1210	CAP-47U-0003	1
GRM1885C1H302JA01D	3000 pF ±5% 50V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)	C69	CAP-0603	CAP-3N-0001	1
860040674004	100 µF 50 V Aluminum Electrolytic Capacitors Radial, Can 7000 Hrs @105°C	C77, C86	WCAP-ATUL_8X11.5(DXL)	CAP-100u-0001	2
BL-HKC36G-AV-TRB	LED YELLOW CLEAR 0603 SMD	D1, D2, D15, D16, D17, D19	LED-0603	LED-0001	6
PMEG6030EP,115	DIODE SCHOTTKY 60V 3A SOD128	D3, D4, D5, D6, D9, D10, D11, D12	DIO-SOD128	DIO-SCHOTTKY-0003	8
SMBJ12CA	TVS DIODE 12VWM 19.9VC DO214AA	D7, D8, D13, D14	DO-214AA (SMBJ)	DIO-TVS-0002	4
PDS760-13	DIODE SCHOTTKY 60V 7A POWERDI5	D18, D20	DO-POWERDI5	DIO-SCHOTTKY-0001	2
SMBJ40CA	TVS DIODE 40VWM 64.5VC DO214AA	D21, D22	DO-214AA (SMBJ)	DIO-TVS-0001	2
0467.250NRHF	FUSE BRD MT 250MA 32VAC 32VDC	F1	RES-0603	FUSE-0002	1
01550900M	FUSE BLOK CARTRIDGE 125V 10A SMD	F2, F3	FUSE-0001	FUSE-0001	2
LM1117IMPX-3.3/NOPB	IC REG LIN 3.3V 800MA SOT223-4	I1	IC-SOT-223-4	IC_LDO_3V3_0001	1
LM1117MPX-5.0/NOPB	IC REG LINEAR 5V 800MA SOT223-4	I2	IC-SOT-223-4	IC_LDO_5V_0001	1
FTSH-107-01-L-DV-K	Connector Header Surface Mount 14 position 0.050" (1.27mm)	J4	SAMTEC_FTSH-107-01-L-DV-K	CONN_Flashing-0001	1
CSAD1010-150M	SMD MOLDING POWER CHOKE	L1	IND-15U-0002	IND-15U-0002-NoJLC	1
IHLP2525CZER100M11	IHLP-2525CZ-11 10 20% ER E3	L2	IND_IHLP_2525_CZ_VIS	IND-10U-0002	1
CSD17575Q3	MOSFET N-CH 30V 60A 8VSON	Q1, Q4, Q5, Q6	IC-8VSONCLIP	NMOS-26A-0001	4
T2N7002AK,LM	MOSFET N-CH 60V 200MA SOT23	Q2, Q3, Q7, Q8	IC-SOT23-3	NMOS-200MA-0001	4
CSD18543Q3AT	MOSFET N-CH 60V 12A/60A 8VSON	Q9, Q10	IC-8VSONCLIP	NMOS-60A-0001	2
ERJ2GEJ1R0X	RES SMD 1 OHM 5% 1/10W 0402	R1, R10	RES-0402	RES-1-0001	2
RESC1607X60N	0 Ohms Jumper Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R2	RESC1607X60N	RES-0-0003	1
RC0603FR-074K7L	RES 4.7K OHM 1% 1/10W 0603	R3, R4	RES-0603	RES-4.7k-0001	2
RC1210FR-07120RL	RES 120 OHM 1% 1/2W 1210	R5	RES-1210	RES-120-0001	1
RC0402FR-07100RL	RES 100 OHM 1% 1/16W 0402	R6, R8, R19, R20	RES-0402	RES-100-0001	4
RC0402FR-07470RL	RES 470 OHM 1% 1/16W 0402	R7, R9	RES-0402	RES-470-0001	2
RC0402FR-7W33RL	RES 33 OHM 1% 1/8W 0402	R11, R12, R13, R14, R15, R16, R17, R18	RES-0402	RES-33-0001	8
RC0402FR-07240RL	240R 0.063W 1% 0402 (1005 Metric) SMD	R21, R29	RES-0402	RES-240-0001	2
RC0402FR-071KL	RES 1K OHM 1% 1/16W 0402	R22, R30	RES-0402	RES-1k-0001	2
RC0603FR-072M15L	RES SMD 2.15M OHM 1% 1/10W 0603	R23, R31	RES-0603	RES-2.15M-0001	2
RC0402FR-07147KL	RES 147K OHM 1% 1/16W 0402	R24	RES-0402	RES-147k-0001	1
RTT021783FTH	62.5mW Thick Film Resistors 50V ±100ppm/℃ ±1% 178kΩ 0402 Chip Resistor - Surface Mount ROHS	R25, R34	RES-0402	RES-178k-0001	2
RC0603FR-07100KL	RES SMD 100K OHM 1% 1/10W 0603	R26, R33, R38, R44	RES-0603	RES-100k-0001	4
RC0402FR-0759KL	RES 59K OHM 1% 1/16W 0402	R27	RES-0402	RES-59k-0001	1
RC0402FR-0710K5L	RES 10.5K OHM 1% 1/16W 0402	R28	RES-0402	RES-10.5k-0001	1
RC0402FR-0780K6L	80.6 kOhms ±1% 0.063W, 1/16W Chip Resistor 0402 (1005 Metric) Moisture Resistant Thick Film	R32	RES-0402	RES-80.6k-0001	1
RC0402FR-0736K5L	36.5 kOhms ±1% 0.063W, 1/16W Chip Resistor 0402 (1005 Metric) Moisture Resistant Thick Film	R35	RES-0402	RES-36.5k-0001	1
RT0805BRD0710K4L	10.4 kOhms ±0.1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Thin Film	R36	RES-0805	RES-10.4k-0001	1
RC0603FR-07200KL	RES SMD 200K OHM 1% 1/10W 0603	R37, R43	RES-0603	RES-200k-0001	2
C337220	2512 Chip Resistor - Surface Mount ROHS	R39, R40, R41	RES-2512	RES-0.5m-0001	3
RC0402FR-0710KL	RES 10K OHM 1% 1/16W 0402	R42	RES-0402	RES-10k-0001	1
UCC27282DRCR	IC GATE DRVR HALF-BRIDGE 10VSON	U1, U4	IC-10VSON	IC-GTDR-0001	2
STM32G431CBT6	IC MCU 32BIT 128KB FLASH 48LQFP	U2	IC-LQFP48	IC-STM32G431CBT6	1
TCAN1044VDRQ1	IC TRANSCEIVER 1/1 8SOIC	U3	IC-8SOIC	IC-CAN-0001	1
SN74LVC2G157DCUR	Multiplexer 1 x 2:1 8-VSSOP	U5, U6	IC-8VSSOP	IC-MUX-0001	2
TPS54561DPRR	IC REG BUCK ADJ 5A 10WSON	U7, U8	IC-10WSON	IC-BUCK-0001	2
Molex - Micro-Fit 3.0 Series connector, 3.00mm (0.118") pitch header, 4 positions.		X1, X3	43045-0400	HEADER_POS4_?43045-0401	2
Molex 43045-0600	Molex 43045-0600	X2, X4	43045-0600	Micro-Fit Molex 6 Pin 2 Row Right Angle	2
Molex 43650-0300	Molex 43650-0300	X5, X6	43650-0300	Micro-Fit Molex 3 Pin Right Angle	2
Molex - Micro-Fit 3.0 Series connector, 3.00mm (0.118") pitch header, 2 positions.		X7, X8, X9	43045-0200	HEADER_POS2_43045-0200	3